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Catching a falling star: Mobility of declining star performers, peer effects, and organizational performance in the National Football League

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ABSTRACT

Research examining how star performers impact their peers largely finds that stars positively impact peer performance. Another stream of research has examined how stars impact their organizations, specifically after mobility events, and finds that hiring stars often leads to lower organizational performance. However, these research streams have developed independently. This paper attempts to reconcile these findings by examining the mobility of stars in the twilight of their career. Leveraging a dataset of individual and organizational performance in the NFL, we find that falling star quarterbacks have positive impacts on their peers when they change organizations. This peer effect leads to increased organizational performance, even though the individual performance of the falling star declines. Moreover, we test and compare the specific mechanisms driving this effect.

"*He's a guy that raises all boats. He's already made (his teammates) better, and they haven't met him yet just because of the type of person he is, his reputation and what he's done in this league. So, he's just going to have a tremendous effect on the Denver Broncos.*"
–*John Elway on the Denver Broncos signing of Peyton Manning*[1]

1. Introduction

The impact that highly talented individuals—i.e., star performers—have on their peers and their organization has long fascinated scholars (Ernst, Leptien, & Vitt, 2000; Lotka, 1926; Narin & Breitzman, 1995; Zuckerman, 1967) and is a topic that has garnered a great deal of attention in this century (Agrawal, McHale, & Oettl, 2014; Azoulay, Graff Zivin, & Wang, 2010; Call, Nyberg, & Thatcher, 2015; Groysberg, Lee, & Nanda, 2008; Groysberg & Lee, 2008; Oettl, 2012). Research has shown that star performers can affect their organizations in multiple ways. First, since stars have high levels of individual ability and human capital, their individual abilities can lead to higher organizational-level human capital, which has been shown to lead to improved organizational performance (Crook, Todd, Combs, Woehr, & Ketchen, 2011). Second, research has shown that star performers can also positively impact the performance of their colleagues (Agrawal et al., 2014; Azoulay et al., 2010; Lacetera, Cockburn, & Henderson, 2004; Oettl, 2012). For example, a positive spillover effect has been found in a diverse set of industry contexts, including retail cosmetic sales (Chan, Li, & Pierce, 2014a), science and academic research (Agrawal et al., 2014; Azoulay et al., 2010; Oettl, 2012), grocery store clerks (Mas & Moretti, 2009), and university students in India (Hasan & Bagde, 2013, 2015).

However, other studies have questioned the conventional wisdom of the value and importance of stars (Chambers, Foulon, Hanfield-Jones, Hankin, & Michaels, 1998), particularly when stars move to new organizations. In an equally diverse set of contexts, researchers have found a decline in star performance after changing organizations (Groysberg et al., 2008; Groysberg & Lee, 2008, 2009; Huckman & Pisano, 2006) and a negative impact on the organization's performance (Berman, Down, & Hill, 2002; Groysberg et al., 2008). The logic underpinning these findings is that when a star performer changes organizations, the star and the organization must overcome significant adjustment costs (Hatch & Dyer, 2004; Hitt, Bierman, Shimizu, & Kochhar, 2001) and that the individual may be leaving behind firm-specific human capital tied to the organization that cannot be leveraged in their new

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[1] https://www.espn.com/nfl/story/_/id/7713078/2012-nfl-free-agency-peyton-manning-denver-broncos-agree-96m-deal.

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environment (Becker, 1993; Groysberg et al., 2008).

While these findings yield important insights, many open questions remain, perhaps because we do not fully understand how these processes impact all three actors (the star performer, the organization, and the star's peers) concomitantly. One specific unanswered question is what happens over time as a star's individual ability begins to fade. This is an important question, since even as stars age, they are still highly sought after in the labor market—yet we do not understand all the ways in which these stars can impact organizations. Call and colleagues state that "when stars reach a career pinnacle and secure their reputations, individual performance is likely to become less important" (Call et al., 2015, p. 634), which leads to the following question: can a so-called 'falling star' still drive positive spillover or peer effects, and if so, why and how? This research seeks to answer these questions about falling stars, peer effects, and organizational performance. Building on influential research that categorizes different *types* of star performers (Oettl, 2012), we argue that an individual star performer's contribution can also be categorized into different 'stages' with respect to performance over time. In other words, if we consider the performance of a star as an intertemporal arc, which begins with a rise early in their career, an apex in the prime of their career, and a subsequent decline, we can begin to address these questions. Gaining a better understanding of the value of this type of star to organizations can help researchers better identify the microfoundations of competitive advantage and organizational performance (R. Coff & Kryscynski, 2011; Felin, Foss, Heimeriks, & Madsen, 2012; Felin & Foss, 2005; Felin & Hesterly, 2007).

Importantly, prior research has shown that not all professions or contexts universally follow this inverted-U shape trajectory regarding performance (Way, Morgan, Clauset, & Larremore, 2017). However, we argue that 'falling stars' are particularly prevalent in highly competitive (either physically or cognitively) professions that are characterized by high pressure or high stress environments. Indeed, it is in these contexts that the (often rapidly) declining performance of star performers should be particularly salient. These include but are not limited to surgeons, investment bankers, air-traffic controllers, Olympic sports, Silicon Valley tech startups, emergency room doctors, enlisted soldiers, gourmet restaurant chefs, professional gamers, sales managers, firefighters, and the like. In all these contexts, the very best invariably reach a point where they are no longer able to perform at the same level they once did, or the same level of those younger, faster, or more desperate to succeed.

There is also a lack of research examining how peer effects potentially impact organizational performance. It is important to examine both effects simultaneously because even if star performers lose some of their firm-specific human capital (Groysberg & Lee, 2009), they could still positively impact their organizations through these peer effects. Failure to examine both could lead researchers to draw the wrong conclusion surrounding the value of star performers, especially those who have already begun to decline in performance.

We posit that a falling star (i.e., a star in the twilight of their career whose performance has begun to fade) can impact their colleagues in two possible ways: through knowledge sharing (Azoulay et al., 2010; Chan et al., 2014a; Oettl, 2012) or complementarities in job design (Ethiraj & Garg, 2012), which we define as the degree of interdependence of specific job roles within the organization. It is important to note that either one or both of these mechanisms could be at work simultaneously. In attempting to tease apart these potential mechanisms, we examine several contingencies with respect to the peers that a star interacts with. Ultimately, stars at different stages of their career may be more or less likely to produce spillover effects, and different types of peers may be more or less receptive to their influence. We find support for our argument that adding a falling star can lead to an increase in new peer performance (even if the star him/herself is not performing better). We also find that these peer effects act as the mechanism by which adding a falling star can positively impact organizational performance.

To test our hypotheses, our study uses data from a professional sports league. Sports data are particularly valuable for this study since they allow us to accurately track the intertemporal arcs of players' careers and easily identify when a star's individual ability begins to decline. Furthermore, even as physical ability declines, both knowledge sharing and complementarities could allow players to continue to contribute to the team. Specifically, we turn to data from the National Football League (NFL) and investigate the impact that star quarterbacks (QBs) have on a new team and teammates after a move in the twilight of their career. The focus is on the star quarterback because this is generally considered the most star-studded position in professional football and arguably one of the most star-studded positions in all professional sports. Leveraging a sample of 36-star quarterback mobility events in the NFL over 22 years, the study leverages a unique dataset that compiles comprehensive team performance measures for every team in the NFL, player Pro-Bowl (PB) awards, and individual player statistics for every player in the NFL since 1993 (the beginning of free agency). Overall, the findings show that gaining a declining star quarterback leads to an increase in the number of teammates invited to the pro bowl (suggesting peer effects are at play in the NFL) and an increase in the number of wins for the team adding the star QB. Importantly, the study finds these effects even though the star's own performance is in decline. Ultimately, our findings contribute to the literature on star performers, labor mobility, and peer effects, specifically in highly competitive and pressure-filled professional settings.

2. Theory and hypothesis development

2.1. Falling stars

Human capital is the knowledge, skills, and ability that an individual possesses (Becker, 1962, 1993). Star performers are highly talented individuals whose skills distinguish them from the average worker (Ernst et al., 2000; Lotka, 1926; Narin & Breitzman, 1995; Oettl, 2012; Zuckerman, 1967). Highly talented star performers are much more productive than average performers (Lotka, 1926; Narin & Breitzman, 1995; Simontin, 1999), which leads one author to conclude that "lesser talent is often a poor substitute for greater talent" (Rosen, 1981, p. 846). More recent studies of stars have examined a broad range of industries and professions, including professional sports (Berman et al., 2002; Ethiraj & Garg, 2012), financial analysts (Groysberg et al., 2008; Groysberg & Lee, 2008, 2009), cardiac surgeons (Huckman & Pisano, 2006), and scientists (Lacetera et al., 2004).

In a reconceptualization of star human capital, Oettl (2012) makes a compelling argument that the impact of stars is not limited to their individual performance and that examining the impact of stars should include the effect they have on colleagues. In their framework, not all stars are the same. Some are focused on their individual performance, but a second type of star is likely to help colleagues. When those in the latter category died, the output quality of their former colleagues declined (Oettl, 2012).

In the same sense that the conceptualization of a star is multidimensional, the arc of an individual star's career can also be dimensionalized into different 'stages' with respect to his or her output or performance (Call et al., 2015). Perhaps the most prototypical arc that one can imagine of a star's career begins with a rise to stardom and continues with the prime of one's career, followed by a downward trend (the slope of which depends on the industry) in productivity at the end of a career. Some evidence exists that career productivity, in general, follows the pattern we describe (Galenson, 2009, 2011; Simontin, 1997), but this research examines all creative workers. To the best of our knowledge, we know of no other study that specifically examines the dimensionality *within a star's career* rather than the dimensionality across different types of stars. Perhaps this is the case because the studies that have looked at peer effects from star performance (Agrawal et al., 2014; Azoulay et al., 2010; Chan, Li, & Pierce, 2014b; Oettl, 2012) have tended to be in contexts where stars can be in the prime of their career for a significant amount of time. For example, in knowledge-intensive

industries such as academia or the sciences, star performers can ostensibly maintain high levels of performance for decades. However, this emphasis overlooks the fact that many industries (even sports industries) require some mix of physical and mental abilities (Chacar & Coff, 2000; Chacar & Hesterly, 2008), and the ratio of one ability to another can be heavily influenced by the specific stage of a star's career.

Admittedly, not all industries or professions appear to universally follow this downward arc in the late stages of a star's career. For example, in a comprehensive analysis of all PhD-granting tenure-track computer science departments in U.S. universities, Way et al. (2017) find that faculty careers did not uniformly follow this inverted U-shaped career trajectory, and the patterns of faculty productivity over time were more diverse than prior research has suggested. However, even in knowledge-intensive industries, career productivity seems to tail off at the end of an individual's career (Levin & Stephan, 1991). Additionally, Groysberg and colleagues (2008) show (through a control variable) that the longer a star financial analyst is ranked, the less likely they will remain ranked, providing some evidence of the performance decline even in knowledge-based industries.

We believe that the distinction hinges on an important scope condition to our theory. Specifically, we expect a more pronounced decline in the performance of star performers later in their careers when they are in professions or industries that are highly competitive and characterized by high pressure or stress environments. For example, in our context (the NFL), star quarterbacks can go from perennial all-stars to fighting to save their jobs with the next crop of promising young talent in a matter of a couple of years because of the physical and mental demands of the job and the extremely competitive nature of the sport. In these contexts, the physical and mental demands are simply higher and more taxing for aging stars. While this context is unique in that the intertemporal arc of a star quarterback's career is usually compressed into a decade or so, rather than several decades in other professions, we expect our theory related to falling stars and performance to generalize to other similarly competitive, high-pressure environments.

In contrast, in more stable and less pressure-packed professions, we would still expect to see signs of the same performance arc for stars, but output across individual stars should be more diverse. For example, in the context of academia and computer science departments mentioned above (Way et al., 2017), the key distinction with a tenure-track faculty position is the enormous job security that exists post-tenure. In other words, distinguished computer science professors are not typically driven out of their positions by young up and coming junior professors. This lack of direct labor competition and pressure should lead to more diverse and idiosyncratic productivity rates across senior faculty over time. Some will naturally become less productive because job security does not hinge entirely on productive output, while others will remain highly productive because this same job security reduces burnout and stress.

Overall, in hypercompetitive industries, we believe that star performers will predominantly follow the same career performance arc, including a drop in productivity at the end of their careers.

H1: In highly competitive, high stress professions, a star performer's career shows a nonlinear relationship with performance such that there is an early period of increasing performance, a middle period of relatively constant performance, and a later period of declining performance.

Next, this paper follows influential research (Groysberg et al., 2008; Groysberg & Lee, 2008, 2009) that argues that a star performer's individual skills decrease when they change organizations. The general explanation for postmobility performance decline for stars is that there is a strong firm-specific component that contributes to their superior performance (Becker, 1962, 1993). Even in cases that, at least seemingly, have a high level of general human capital, a lack of requisite firm-specific knowledge may limit an individual's ability to transfer skills to another firm (Groysberg et al., 2008). This firm-specific component leads to adjustment costs while the newly hired star develops tacit, organizational knowledge (Hitt et al., 2001; Hatch & Dyer, 2004). This adjustment cost should occur in all stages of a star's career, and therefore, a falling star should also exhibit a decrease in individual productivity when they change organizations. However, this research typically only examines the impact of the star's own individual performance upon changing organizations and not their ability to produce a spillover effect in their new peers.

2.2. Falling stars and spillover effects

If organizations are more likely to hire a star in the twilight of their career who have already begun to show a decline in performance, and by changing teams, this performance decline gets worse, this leads to an interesting question: why bring these stars into the organization? One potential answer is that star performers bring more to the organization than simply their individual skills and overall performance.

We detail two likely mechanisms by which falling stars could positively impact their new peers. The first mechanism that may allow a falling star to positively influence peers after a move is through knowledge sharing. When a star moves to a new organization, even if they leave behind some degree of firm-specific human capital in their old organization (Groysberg et al., 2008), it is also likely that these stars bring with them some extant reserve of knowledge that they can ideally impart on their new set of peers to positively influence performance (Lacetera et al., 2004). This can include mentoring younger colleagues, teaching those around them how to work harder or more efficiently, and sharing better routines or best practices that can raise levels of performance (Zuckerman, 1967). For example, in our empirical context, star QBs at the later stages of their career often change teams with the clear intention to compete for a starting job on the team. Often, however, the team envisions the falling star mentoring young up and coming stars who are currently on the team.

For example, when Matt Hasselback moved to the Indianapolis Colts in 2013, there was already a clear rising star quarterback on the team (Andrew Luck), so Hasselback knew that mentoring would be a key component of the job description. Indeed, according to an article written at the time, "Working with Luck has been a joy, even if Hasselbeck's thrown only three passes this season and likely won't throw many more. The reward has indeed been giving everything he has to a player who has the job he would love to have again."[2] Nevertheless, during the 2015 season, Hasselback was also pressed into action for 4 games (25 % of the season) when Andrew Luck was sidelined with an injury. He proceeded to lead the team to wins in all four games, finishing the season with a 4–0 record as a starter, which was the first time a quarterback over the age of 40 had accomplished this.

A falling star likely has an advantage over younger stars in regard to knowledge accumulation since they have had more time to develop a general knowledge of their industry, competitors, and strategies employed by organizations that could be useful to a variety of organizations (Becker, 1993). This higher level of accumulated general knowledge gives them the ability to materially impact the performance of those around them through knowledge sharing. Moreover, these knowledge-sharing mechanisms may be particularly important in knowledge-intensive industries. Even in job designs that are driven to a large degree by routine behaviors or practices, such as retail sales, (Chan et al., 2014a), Chan and colleagues show that knowledge transfer plays a primary role. Furthermore, even in job designs that require a broader mix of abilities, the role of knowledge sharing should be important. As a star whose job requires a mix of physical and mental skills reaches the twilight of their career, physical abilities decline at a faster rate than their mental abilities and knowledge (Moir, 2009). Therefore, even if the

[2] https://sports.yahoo.com/news/nfl–matt-hasselbeck-s-mentoring-of-andrew-luck-includes-a-rigged-contest-and-rubbing-people-the-wrong-way-010528025.html.

star performs at a lower level individually (based on both the loss of physical abilities and loss of prior organization-specific knowledge), they are still likely to have a high level of general knowledge that they can share with their peers.

An additional potential mechanism that could drive peer effects of falling stars is through complementarities in ability or job design. Similar to the dimensionality of stars discussed above, the individuals who make up an organization can also be dimensionalized across several criteria. For example, a star performer may move to a new organization yet rarely interact with some peers in other departments. Thus, the relative distance between job positions may play a role, and we would expect that those that work or collaborate (Agrawal et al., 2014; Azoulay et al., 2010) directly with stars would stand to benefit the most. Peers whose goals match the skills of a star they work with may also stand to benefit more (Hasan & Bagde, 2013). Furthermore, peers that share similar levels of experience or complementarities in job design may stand to benefit more from stars coming into an organization. For example, when Michael Vick joined the Philadelphia Eagles in 2009, he was quoted as saying, "I'm always going to be a leader on this team, regardless of what my place is. If I have to watch film with all the quarterbacks, or if I have to coach the quarterbacks one day, then that's what I'll do."[3].

This sort of symmetric interdependence (Ethiraj & Garg, 2012) suggests that the ability of the star to lift the performance of those around them may be contingent not only on the complementarity between the star and a peer but also on the complementarity of the job design itself. Interestingly, peers who enter the organization along with the star or subsequent to the star's arrival (Agrawal et al., 2014) may also benefit more as well. This could be attributed to the fact that new entrants, whether stars or not, share a common bond or experience of having to learn a new organizational environment or culture. They may also be less susceptible to inertia or path dependence in terms of existing norms or routines within the firm. Furthermore, the peer effects from these complementarities could increase over time as the average organizational tenure increases, although there may be diminishing returns to these increases as this tenure continues to rise (Subramony & Chadwick, 2021). Finally, in contrast to knowledge sharing, we would expect these processes to manifest more prominently in industries or job designs that require physical abilities as well as knowledge-intensive activity. Ultimately, complementarities may drive peer effects, particularly in job designs that require some degree of symmetric interdependence. These complementarities should also exist for falling stars. Even if the star's physical abilities are in decline, they still may have more abilities than other individuals, and their role may better complement some peers. Therefore, if the falling star complements a peer's performance, these peers are still likely to perform at a higher level.

Aging stars are often sought by organizations for myriad reasons, both tangible and intangible, and therefore, we have argued that whether a star performer can significantly improve peer performance is a particularly interesting question for falling stars, which are in the twilight of their career. In addition to how a falling star could impact their peers, it is important to note that falling stars, despite a reduction in their individual performance, are still likely to perform better than an average worker. This higher level of performance could also positively impact both the new organization and the falling star's new peers. While falling stars may be the best equipped to help others because of their experience and knowledge, they may also complement their new peers and lead to an increase in peers' performance. These statements lead to our second hypothesis:

H2: Adding a falling star to an organization is associated with an increase in the star's peers' performance.

2.3. Falling stars and organizational performance

While conventional wisdom suggests that adding a star performer will lead to increases in organizational performance (Chambers et al., 1998; Crook et al., 2011), influential research has suggested that even star performers with highly general skills possess a firm-specific component that may limit their ability to positively impact their new organization once they join it (Berman et al., 2002; Huckman & Pisano, 2006; Groysberg et al., 2008; Groysberg & Lee, 2008, 2009). This firm-specific component should be much more valuable in the organization in which it was created (Becker, 1993). Therefore, individual skills, even if thought to be general, are less transferable across organizational contexts. This should be truer for falling stars, who have already begun to show a decline in individual performance, and changing organizations should exacerbate this decline. These arguments have found empirical support, and studies suggest that adding a star performer leads to decreases in organizational performance (Groysberg et al., 2008; Groysberg & Lee, 2008, 2009). However, these studies do not account for the possibility of peer effects that increase the level of performance of an added star's new colleagues, and this increase in colleague performance could positively impact an organization's performance.

Above, we have argued that falling stars can positively impact their new colleagues in two potential ways: through knowledge sharing or through complementarities in job design (or some combination of both). While prior research has shown positive peer effects from a star performer (Azoulay et al., 2010; Ethiraj & Garg, 2012; Lacetera et al., 2004; Oettl, 2012), none of these studies have examined the organizational performance implications of these peer effects. We argue that when an organization hires a star, even a falling star, the organization's current employees' individual performance increases through the star's ability to drive peer effects. Essentially, we are arguing that the star's new colleagues' human capital (their knowledge, skills, and abilities) increases, leading to an overall increase in the human capital of the organization. Additionally, it is likely that the falling star's own abilities are higher than the employee they replace despite being reduced from their peak, also potentially leading to an increase in the organization's human capital. This overall increase in human capital has been found to be positively related to an increase in organizational performance (Crook et al., 2011; Hitt et al., 2001). Based on this and our arguments from above, adding a falling star should positively impact the performance of their new organization. Therefore, we hypothesize the following:

H3: Adding a falling star to an organization is associated with an increase in the performance of that organization.

3. Methods

3.1. Data and sample

This paper examines the movement of star players in the National Football League (NFL). As stated earlier, the NFL is an appropriate context to test our hypotheses. First, the abundance of data on individual-level performance allows us to easily determine the intertemporal arc of each player's career. This will allow us to determine which players are star performers and when these star performers' physical abilities start to decline. Furthermore, this setting has easily discernable player movement between organizations, which allows us to track the mobility of falling stars. The setting also requires a mix of skills, including physical and mental abilities (Chacar & Coff, 2000; Chacar & Hesterly, 2008; Gould, 2000), which can help us separate out knowledge sharing and complementarities when the star's physical ability has

[3] https://bleacherreport.com/articles/1815872-michael-vick-setting-an-example-for-the-philadelphia-eagles-with-his-leadership.

begun to fade. Finally, there are also abundant organizational-level performance measures that can help us access how falling stars have impacted the overall organization.

The NFL was founded in 1920 and has become the most watched sport in the U.S., increasing the popularity of both teams and individual players. Currently, there are 32 teams in the league, and each play a 16-game regular season (during our data window), with the highest performing teams then competing in the postseason to win the NFL championship (the Super Bowl). Each NFL roster has 53 active players, with more on practice squads (who compete only in practice but do not participate in games) and injured reserves (players who are hurt and miss significant time). In each season, there are over 1600 players on active rosters throughout the league. Additionally, football is a competitive team sport, with individual players on different teams performing similar, interdependent tasks (Aime, Johnson, Ridge, & Hill, 2010; Holcomb, Holmes, & Connelly, 2009). There are three phases to the game: offense (when the team has control of the ball), defense (when the opposing team has control of the ball) and special teams (which includes kickoffs, field goals, and punting). Each of these phases has specialized players and coaches. While each phase operates separately during the game, they spend the week preparing for the game together in practice. Therefore, while offensive players are only on the field together during a game, they spend significant time with their peers who participate in other phases.

While recent research has looked at labor mobility in the NFL, specifically hiring decisions related to coaches (Rider, Wade, Swaminathan, & Schwab, 2016), we know of no other study that has looked at labor mobility and star performance in the NFL. This study isolates the differential effect of star quarterbacks (QB), who are in the later stages of their careers. These players are generally considered to have the most impact on their team's success or failure, since when the team is on offense, the QB handles the ball on every play. Teams can win games with poor performing QBs, but largely, the more successful the QB, the greater the team wins. The worst teams in each year get to pick first in the draft (of college players) the following year, and since 1993, more QBs have been drafted first overall than any other position (14 out of 23)[4], which underscores the importance of the position. Finally, QBs tend to be the face of the franchise, with fans and commentators often discussing a team as "the QB's team".

Due to changes in the institutional context of the NFL (e.g., rule changes), this study has chosen to examine the NFL only in the free agency period, beginning in 1993. Prior to 1993, the only way a player could change teams was if the organization traded them for others. After 1993, individuals who were not under contract can negotiate with and move to other organizations who desire their service players. This latter situation more closely matches the labor market in other industries, in which individuals and organizations can try and match skills and needs for a given position. Data collection stopped in 2015. While data on the NFL are available from a variety of sources, the data for this study were collected largely from two online websites, https://www.pro-football-reference.com and https://www.nfl.com. The data collection started in 1992. This allowed for the lagging or differencing of organizational and individual performance variables in order to begin to examine the performance implications of star QB mobility from 1993 to 2015.

3.2. Star measure

To model how star QBs drive spillover effects after a move to a new team, we must first identify which QBs are considered stars. We define a star player as one that has been voted on three or more NFL Pro Bowls. A list of all QBs that fit these criteria is given in Table 1. The Pro Bowl is the NFL's version of an All-Star game, where the best players in the league that year play an exhibition game against one another. Players are selected for the Pro Bowl by a combination of fans' votes, coaches' votes, and votes from other players[5]. Thus, we believe that this captures a composite of star qualities (reputation and popularity, technical ability, and competitive performance). Many player contracts also include incentives for making the Pro Bowl[6], making it a goal that many players strive to achieve. Additionally, players are often introduced and discussed in the media as Pro Bowlers with more Pro Bowl appearances indicating a higher-level performer. For example, Aaron Rodgers, a quarterback with the Green Bay Packers, is often introduced today as a "5-Time Pro Bowler". This study chose to use 3 Pro Bowls as the measure (as opposed to a single Pro Bowl selection) to overcome any possibility that an individual had only one strong season. This ensured that the study was truly capturing star performance and not an individual who had one strong season and who was unable to repeat their strong performance (so called one-hit wonders). While many players choose not to play in the Pro Bowl if selected (due to prior injuries, concerns for future injuries, playing in the Super Bowl, etc.), we code a player as a Pro Bowler if they were selected for the team, even if they chose not to participate. Alternates who chose to participate were also coded as being Pro Bowl players in that year.

Table 2 shows statistical tests comparing 3-Pro-Bowl (3-PB) star QBs

Table 1
List of star quarterbacks.

Name	*Active Player*	*Mobility Count*	*Total Pro-Bowls*	*Total All-Pros*	*Hall of Fame*
*Aaron Rodgers**	*Yes*	*0*	5	2	*No*
*Ben Roethlisberger**	*Yes*	*0*	4	0	*No*
Boomer Esiason	*No*	*3*	4	1	*No*
Brett Favre	*No*	*2*	11	3	*Yes*
Carson Palmer	*No*	*0*	3	0	*No*
Dan Marino	*No*	*0*	8	3	*Yes*
Dave Krieg	*No*	*4*	3	0	*No*
Donovan McNabb	*No*	*2*	6	0	*No*
Drew Bledsoe	*No*	*2*	4	0	*No*
*Drew Brees**	*No*	*0*	9	1	*No*
Eli Manning	*No*	*0*	4	0	*No*
Jeff Garcia	*No*	*4*	4	0	*No*
Jim Kelly	*No*	*0*	5	1	*Yes*
Joe Montana	*No*	*1*	7	3	*Yes*
John Elway	*No*	*0*	9	0	*Yes*
Kurt Warner	*No*	*2*	4	2	*No*
Mark Brunell	*No*	*3*	3	0	*No*
Matt Hasselbeck	*No*	*2*	3	0	*No*
Matt Ryan	*Yes*	*0*	3	0	*No*
Michael Vick	*No*	*3*	4	0	*No*
*Peyton Manning**	*Yes*	*1*	14	7	*Yes*
Philip Rivers	*No*	*0*	5	0	*No*
Randall Cunningham	*No*	*3*	4	1	*No*
Rich Gannon	*No*	*0*	4	2	*No*
Steve McNair	*No*	*1*	3	0	*No*
Steve Young	*No*	*0*	7	3	*Yes*
*Tom Brady**	*Yes*	*0*	11	2	*No*
Tony Romo	*No*	*0*	4	0	*No*
Troy Aikman	*No*	*0*	6	0	*Yes*
Warren Moon	*No*	*3*	9	0	*Yes*

Note: Mobility count only considers moves after player plays in 3rd Pro-Bowl. Active player shows if the player was active at the end of our data collection period.

* These players were not yet eligible for the Hall of Fame at the end of our data window but are seen as highly likely to get in once they become eligible.

[4] https://www.nfl.com/draft/history/alltimeno1.

[5] https://www.nfl.com/probowl.

[6] https://sports.yahoo.com/blogs/nfl-shutdown-corner/champ-bailey-julius-peppers-top-list-nfl-pro-163029343–nfl.html.

Table 2
Comparison of star quarterbacks with good quarterback (those who have played in 1 or 2 career Pro-Bowls).

Variable	*Good players*	*Star players*	*Difference*	*t-statistic*
Career length (yrs)	11.63	13.72	−2.09	−9.64
Completion percentage	54.58	58.21	−3.64	−3.90
Yards per year	1889.70	2869.85	−980.14	−10.81
TDs per year	11.01	18.84	−7.82	−11.96
QB rating	73.16	82.74	−9.58	−6.48
Ave. ranking	30.07	20.01	10.07	8.98
All-Pro teams (total per player)	0.02	1.03	−13.00	−4.36
Hall of famer (percentage)	0.00	0.29	−0.29	−14.61

Note: Good players defines as those with 1 or 2 career Pro-Bowls. We found 47 players meeting this criterion. We found 30 Star players, with 3 or more career Pro-Bowls, in our data.

to those who played in 1 or 2 Pro Bowls in their career (whom we label as good players). We found 30 star players and 47 good players meeting our criteria out of a total of 340 QBs. As shown, the 3-PB stars perform at a much higher level than those who are considered good players. Furthermore, since 1993, no QB who has played in 2 or fewer Pro Bowls has made the Hall of Fame (HOF) in the modern era, which is the ultimate accolade for a star player. Of those in the sample of 3-PB stars, 26 % went on the make the Hall of Fame. This percentage increases to 46 % if we consider players who have recently retired or are still playing but are widely considered sure bets to make the HOF once they do retire (i.e., Peyton Manning, Tom Brady, Drew Brees, Aaron Rodgers, and Ben Roethlisberger). Alternative measures of star QBs (i.e., 4-time pro-bowlers or those selected to the all-pro team) were not used due to the low number of mobility events, making statistical analysis difficult. Of our star QBs, 15 changed teams for a total of 36 mobility events (some star QBs moved more than once) from 1993 to 2015. Furthermore, a mobility event was only counted for those stars after they made their 3rd Pro-Bowl, since prior to that, it was not certain that the player would become a star.

3.3. Dependent variables

Our first hypothesis examines the career trajectory and mobility of individual star QBs. To assess the star QB's performance, we utilize a comparison-based measure (Call et al., 2015). We calculate a rank based on the 3 main statistical measures of QBs: yards, touchdowns, and completion percentage. For each year, we rank each QB on each variable, with '1′ being the best in the league that season. We then calculate a composite rank as the average of the 3 individual ranks.

To assess the spillover effects at their broadest level, we begin with an examination of both the organization's current members and the organization's performance after the arrival of a star QB in the twilight of their career. First, we calculate the total number of players selected to the Pro Bowl per team in each year (subtracting 1 if the star QB in question was one of those players selected). While many players who are elected do not actually play in the game, we still include them as the measure of a star is the selection based on performance during the season (rather than performance in the Pro Bowl itself). From this, an additional variable was coded to examine only offensive Pro Bowlers, because offensive players have the most direct interaction with the star QB. For each of these variables, change scores were calculated by taking the difference in the number of overall Pro Bowlers (and offensive Pro Bowlers) over the prior year. Here, a higher number signifies an increase in Pro Bowlers for a given team in each year.

Additionally, we augment this with several different measures of organizational performance. First, the study examined the change in wins for a team after the arrival of the star QB. This was calculated as the difference in the number of wins in each year and the prior year. Second, we calculate measures for offensive and defensive rank. While the study is examining the performance implications of adding a star QB, it was not clear, ex ante, whether the added star QB has impact on both the offense and defense, although through a complementarity argument we would expect them to have more impact on the offensive side. To calculate these rankings, we averaged the teams' performance on 2 team-level statistics. For offense, points scored, and total yards gained were the relevant statistics, while for defense, it was points allowed and yards given up. For each of these statistics, a ranking was calculated for each year. The highest ranking is set to 1, and ties are given half-ranks. For example, the highest scoring team in each year would be given a ranking of 1 for points scored, and if 2 teams tied for the most points scored, they would each be given a rank of 1.5. Thus, the overall offensive and defensive ranks represent the average of the 2 relevant statistical rankings for each team for each year. In this instance, the more negative a number is, the better a team's ranking.

3.4. Independent and control variables

For H1, the main independent variable was the number of years of NFL experience for a given QB at the beginning of each season. This measure started with '0′ for rookies and ranged to 20 years. We also calculate the square of this measure to test for the hypothesized nonlinear effects. We also add a binary predictor indicating a mobility event, with '1′ indicating the QB changed teams that year. To test if stars were different from other players, we also included categorical indicators of star status. This is broken into 3 categories: an average player (0 career Pro Bowls), a good player (1–2 career Pro Bowls), and a star player (3 or more career Pro Bowls, as described above).

To assess the spillover effects at the organizational level, we utilize several different control variables in our matching models: lagged offensive rank, lag defensive rank, lag wins, team expected to improve, coaching tenure, coaching change, and year. The first three of these were calculated as described above but lagged to control for the team's performance in the prior season. The *team expected to improve* variable is a binary variable based on the Pythagorean Wins theorem, developed by Bill James (1985) and modified for the NFL[7]. After calculating a team's Pythagorean wins for each season, this is then compared to the team's actual wins. Teams that underperformed their Pythagorean wins are considered better than their record indicates and are more likely to improve in the next season. We calculate this for each year and create a binary variable with '1′ indicating teams likely to improve. Coaching tenure is calculated as the number of years the current coach has been with the team, and coaching change is a binary variable indicating that the coach was either new as the season started or was replaced during the season.

3.5. Identification strategies

To examine the career pattern of our star performers, we use both random and fixed-effects regression models, including both team- and year-level fixed effects and robust standard errors (Wooldridge, 2010). We report results of the random-effects models, since the star category variables do not change over time, and we can therefore not isolate and compare the careers of stars versus other players with fixed-effects models. We also run split-sample fixed effects models on stars only, which show substantively similar results to the random-effects specifications. These results are available from the authors upon request.

To assess the impact of a star's mobility on their new colleagues and teams, we must turn away from traditional panel models. Star players are not randomly assigned when they change teams, and therefore, it is possible that there is some observable or unobservable difference that

[7] https://www.espn.com/nfl/story/_/page/BarnwellNFLPreview160726/predicting-which-nfl-teams-rise-decline-2016-dallas-cowboys-san-diego-chargers.

affects both the propensity for a team to add a star QB and the team's improvement. Traditional panel-data models do not account for this type of endogeneity. Therefore, this study has chosen to utilize a matched sample research design, comparing teams that added a star QB to similar teams that did not add star QB to estimate the average causal effects. Matching methods were chosen to best approximate a randomized experiment with our observational data (Imbens, 2004; Rubin, 1974; Rubin & Thomas, 1996; Wooldridge, 2010). Since it cannot be observed how a team that added a star QB would have improved (or not) had they not added the star QB, matching methods allow for the comparison of teams that add a star QB to those that performed in a similar manner in the prior year yet chose not to add a star QB. While this specification does not control for the possibility of all unobservable differences between teams, it allows for the reduction of bias associated with observable covariates (Stuart, 2010). The study uses nearest-neighbor matching methods with the Mahalanobis distance metric (Stuart, 2010) to predict the average treatment effect, where the treatment is a binary indicator of adding a star QB. Since the study also includes multiple continuous predictors, bias-adjusted robust errors developed by Abadie and Imbens (2011) were used for the continuous predictors. For the organizational level models, matching was performed on 7 control variables: lagged offensive rank, lagged defensive rank, lagged wins, team expected to improve, coaching tenure, coaching change, and year. Checks to ensure that the matches were balanced were also conducted, and the results show that all models were properly balanced (Rubin, 1974; Stuart, 2010).

4. Results

Table 3 shows the individual star-level summary statistics and correlations for testing H1. Table 4 presents summary statistics and correlations for our organizational level models used for testing H2 and H3.

Table 5 shows the results of the random-effects models used to examine players' productivity across their careers. Model 1 shows the relationship between experience and performance for all players, Model 2 breaks this into player categories (stars, good players, and average players, with average players the reference category), and Model 3 adds the mobility effects. Overall, the results suggest a U-shaped relationship between experience and yearly ranking for all players. Furthermore, star players are ranked much lower than other player types ($b = -25.85$, $p = 0.00$), although the pattern is the same across player types. Marginal effects (for NFL experience and star level, with all other variables at their means) with 95 % confidence intervals for these results are shown in Fig. 1 to better understand the magnitude of these effects for each player type category. As shown, star players perform much better than both average and good players, but star players still exhibit a rise/middle/fall pattern in their careers, lending support for H1. Furthermore, Model 3 (Table 5) shows that changing organizations leads to lower performance (on average, changing teams makes these players perform 6.59 ranks lower) and that this effect hurts all players regardless of their ability. This result confirms prior research on star mobility, which shows that even star performers have an organization-specific component to their human capital that requires adjustment costs to overcome when they change organizations (Groysberg et al., 2008).

Our main results for H2 are shown in Table 6. Each of these models estimates the average treatment effect. Model 1 examines the change in Pro Bowlers when adding a star QB. As shown, there is no significant relationship between adding a star and more of the star's new colleagues making the Pro Bowl. However, after isolating the offensive players on the team, a significant spillover effect emerges. Model 2 tests the change in offensive Pro Bowlers by adding a star QB, and the results show that adding a star QB significantly increases the number of offensive Pro Bowl selections for a team, with the results indicating that by adding a star QB, 0.53 more offensive players make the Pro Bowl (95 % confidence interval from 0.06 to 1.01). This provides support for Hypothesis 2—the colleagues of the added star QB appear to be getting better.

Our tests for H3 are shown in Table 7. Model 1 shows a team's change in wins when adding a star QB, and the results show that by adding a star QB, the team gets better, winning on average 0.89 more games (95 % confidence interval ranging from 0.07 to 1.72) in the year following the move. This is a large effect in the NFL, as 1 more win in a season can mean the difference between making the playoffs or not. While making the playoffs is what all players strive for, it also means higher revenue for the team's owner (through more ticket sales, more merchandising sales, more TV revenue, etc.). Model 2 and Model 3 show the change in offensive and defensive rankings, respectively. The results show that a team's offense gets significantly better (ranking increased by −4.28, 95 % confidence interval from −1.62 to −6.94) when they add a star QB, but their defense does not significantly improve. Overall, these last three models provide support for Hypothesis 3 by showing that adding a falling star QB to an organization leads, on average, to an increase in the new organization's performance.

4.1. Additional analysis

To further disentangle the potential mechanisms for the spillover effect, we ran several additional analyses. First, we collected qualitative evidence from our context to assess what coaches, managers, and players thought about adding falling star QBs to their team. We found articles (shown in Table 8, with representative quotes) for all our falling star QBs (except for Joe Montana) discussing what the team felt the QB was bringing to their organization. For example, when the Kansas City Chiefs added Warren Moon, Pro Football History quoted head coach Gunther Cunningham on Moon: "I think he'll be very helpful to Elvis Grbac in his development … I think the players will respond well to him because I think he carries respect throughout the league."[8] As this quote, and many of the other quotes suggest, many teams argued for bringing in a falling star QB in a mentoring role. The organizations respected not only the talent these falling stars showed on the field but also their leadership both on and off the field. These articles suggest that most organizations believe they are bringing in falling star QBs to enhance knowledge sharing within their teams rather than strictly as complementarities.

Second, we ran additional analyses (available from the authors) to better attempt to disentangle the possible mechanisms. For these, we split the mobility events based on the percentage of games the added star plays with the new team (split at 50 %) and ran nearest neighbor matching models. These models, available from the author, show that the performance effect we see is being driven by quarterbacks who play in 50 % or more of the team's games rather than those who play in fewer. This suggests that the mechanism seen in this context is due to complementarities. Another model, showing that adding a star QB increases the number of offensive Pro Bowlers, could also be evidence of an attention-based mechanism, meaning that the star brings attention to their new teammates, and this attention leads them to make more PBs. Further research along this line of inquiry would greatly enhance the understanding of reputation effects and peer performance.

We also ran nearest neighbor matching models on the QBs' individual peers, separated by positions (wide receivers, running backs, defensive players, and kickers)[9], and to conserve space, these models are available from the authors upon request. We found that both wide

[8] https://www.profootballhistory.com/warren-moon/.

[9] Each player was ranked on various important position-specific metrics in each year, and nearest-neighbor matching models were run to see if these players improved ranks or made more Pro Bowls when a star QB was added.

Table 3
Summary statistics and correlation for individual star measures.

	Variable	*N*	*Mean*	*S.D.*	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*
(1)	*Average Ranking*	1985	43.91	22.35	1				
(2)	*Star Level*	1985	0.53	0.76	−0.54*	1			
(3)	*NFL Experience (yrs)*	1985	4.95	4.13	−0.22*	0.39*	1		
(4)	*NFL Exp. Squared*	1985	41.55	57.10	−0.17*	0.38*	0.94*	1	
(5)	*Mobility Event*	1985	0.25	0.43	0.15*	−0.09*	0.25*	0.18*	1

* indicates correlation significant at p <.05 level.

Table 4
Summary statistics and correlation for organizational-level models.

	Variable	*N*	*Mean*	*S.D.*	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*
(1)	*Win Differential*	713	0.02	3.52	1						
(2)	*Team Expected to Improve*	745	0.45	0.5	0.38*	1					
(3)	*Lag Offensive Rank*	713	16	8.54	0.32*	0.25*	1				
(4)	*Lag Defensive Rank*	713	15.99	8.3	0.32*	0.23*	0.17*	1			
(5)	*Lag Wins*	713	7.99	3.03	−0.59*	−0.47*	−0.68*	−0.61*	1		
(6)	*Coach Tenure*	745	2.98	3.38	−0.14*	−0.04	−0.33*	−0.16*	0.36*	1	
(7)	*Coach Change*	745	0.23	0.42	0.11*	0.22*	0.25*	0.18*	−0.34*	−0.39*	1

* indicates correlation significant at p <.05 level.

receivers and running backs improve ranks, but only if they are 2nd ranked players and only if the added star QB plays in 50 % or more of the team's games. These results indicate that the added star improves the performance of 'B' players more than 'A' or 'C' players[10]. These models offer support for complementarities rather than the knowledge-sharing mechanism, as all players are likely to benefit from higher levels of knowledge, but only those who play with the added star QB and have the requisite skills can gain through complementarities. Overall, the qualitative evidence points toward a knowledge sharing mechanism in our context, while our additional empirical tests point toward the mechanism being complementary in job designs. However, we cannot completely rule out that knowledge sharing is occurring, even in these situations.

Table 5
Random effects regression models for average QB ranking versus career length.

Variables	Model 1	Model 2	Model 3
NFL Experience (yrs)	−2.80	−1.49	−2.33
	(0.42)	(0.53)	(0.52)
	[0.000]	[0.005]	[0.000]
NFL Experience Squared (yrs)	0.18	0.13	0.18
	(0.03)	(0.04)	(0.04)
	[0.000]	[0.003]	[0.000]
Good Player		−17.11	−17.22
		(3.94)	(3.87)
		[0.000]	[0.000]
Star Player		−25.48	−25.85
		(4.50)	(4.44)
		[0.000]	[0.000]
Good Player * NFL Experience (yrs)		−1.03	−0.47
		(1.06)	(1.11)
		[0.331]	[0.675]
Good Player * NFL Experience Squared (yrs)		0.06	0.01
		(0.07)	(0.07)
		[0.432]	[0.874]
Star Player * NFL Experience (yrs)		−2.78	−2.06
		(1.28)	(1.26)
		[0.030]	[0.102]
Star Player * NFL Experience Squared (yrs)		0.16	0.11
		(0.09)	(0.09)
		[0.067]	[0.197]
Mobility Event			6.59
			(1.29)
			[0.000]
Good Player * Mobility Event			−1.63
			(2.74)
			[0.660]
Star Player * Mobility Event			1.98
			(2.78)
			[0.481]
Team Fixed Effects	*Included*	*Included*	*Included*
Year Fixed Effects	*Included*	*Included*	*Included*
Constant	48.28	55.25	54.97
	(3.34)	(0.31)	(3.13)
	[0.000]	[0.000]	[0.000]
Observations	1985	1985	1985
Number of Clusters	339	339	339
R-squared (overall)	0.05	0.35	0.37

Robust standard errors in parentheses, p-values in brackets.

4.2. Robustness checks

Our first robustness test was to reduce the threshold for our star measure to see if we were truly capturing star performers. We ran additional matching models, with the same controls as our main models, changing the star threshold from 3 Pro Bowls to 2 Pro Bowls. In these models, available from the author upon request, we find no evidence that adding these players significantly improves team performance. We also run a model to rule out reverse causality, which would indicate that stars are moving to teams that are already expected to improve, and therefore, we would be overestimating any effect of adding a star QB[11]. As shown in Fig. 2, teams expected to decline do show a decline in performance, but when they add a star QB, this decline goes away, for an average difference of 1.20 wins. Teams expected to improve do, on average, improve, and adding a star QB does lead to a small improvement.

5. Discussion and conclusion

In investigating star performers in the NFL, we find that star QBs exhibit similar career patterns to nonstar QBs, and although they perform at a much higher level, they still begin to exhibit a fall-off in performance in the twilight of their careers. Do these falling stars still

[10] These 'B' receivers are players that are still very good but may not yet have reached their full potential. In contrast, first tier 'A' players have likely reached their potential, and third or fourth tier 'C' players simply do not have the requisite skills or playing time to benefit as much from the help a star QB could give.

[11] To test this, we utilize our binary *team expected to improve* variable, based on the Pythagorean Wins theorem described in Section 3.4.

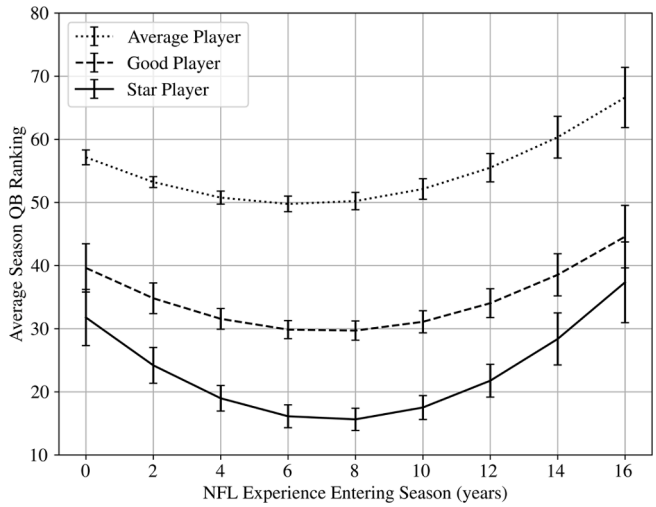

Fig. 1. Marginal effects (and 95 % confidence intervals) of average QB ranking versus NFL experience (years) for average players, good players, and star players.

Table 6
Nearest-neighbor matching models for organizational spillover models.

	Model 1	Model 2
Dependent Variable:	***Change in Pber***	***Change in Off. Pbers***
Average Treatment Effect	0.337	0.537
Robust Standard Error	(0.40)	(0.24)
Z-Statistic	0.83	2.21
Confidence Interval	[-0.45,1.13]	[0.06,1.01]
Total Observations	713	713
Treated Cases	36	36
Number of Matching Variables	7	7
Year Matching	Yes	Yes

Note: Matching variables: lag offensive rank, lag defensive rank, lag wins, team expected to improve, coaching tenure, coaching change, and year.
Robust standard errors bias adjusted for multiple continuous measures.

have positive impacts when they change organizations? Indeed, we find that falling star QBs do have a positive impact when they change organizations, both on their new peers and on the performance of the organizations they join. Adding a star QB leads to more offensive players making the PB, a better ranking on offense, and more wins. These empirical results suggest that the mechanism by which a star QB impacts their new teammates is by increasing complementarities, although the close working relationship between these types of players does not preclude the potential of increased knowledge sharing.

Despite the organizational level model suggesting that the star could bring more attention to their new teammates (increasing the number of offensive PBs), this study found no further evidence of an attention-based mechanism at play in this context. The results from the individual-level models found no evidence that the added star QB's new colleagues (receivers, running backs, and defensive players) make more Pro Bowls, which is found in the organizational-level models. The likely explanation for this is that it is the offensive lineman who is making more Pro Bowls because of the star player moving to their team. Since there are limited statistics for the performance of these players, it is likely that they are judged on other factors, including QB and running back performance. Therefore, adding a star QB could bring more attention to these players, increasing their chances of playing in the Pro Bowl. Unfortunately, due to this lack of statistics on their performance, this explanation cannot be tested.

This study adds to several different research streams. First, this study adds to our knowledge of stars (Agrawal et al., 2014; Azoulay et al., 2010; Oettl, 2012; Rosen, 1981; Zuckerman, 1967) by examining "how stars provide unique organizational value at different stages of their

Table 7
Nearest-neighbor matching models for organizational performance models.

	Model 1	Model 2	Model 3
Dependent Variable:	***Change in Wins***	***Change in Off. Rank***	***Change in Def. Rank***
Average Treatment Effect	0.892	−4.278	−1.729
Robust Standard Error	(0.42)	(1.36)	(1.42)
Z-Statistic	2.13	−3.15	−1.22
Confidence Interval	[0.07,1.72]	[-6.94,-1.62]	[-4.52,1. 06]
Total Observations	713	713	713
Treated Cases	36	36	36
Number of Matching Variables	7	7	7
Year Matching	Yes	Yes	Yes

Note: Matching variables: lag offensive rank, lag defensive rank, lag wins, team expected to improve, coaching tenure, coaching change, and year.
Robust standard errors bias adjusted for multiple continuous measures.

stardom" (Call et al., 2015, p. 634). Specifically, we look at a less examined type of individual star performer—one in the twilight of their career whose skills have begun to diminish. To our knowledge, this is the first study to specifically examine these types of stars. We argued that a star performer's career arc can also be dimensionalized into different 'stages' with respect to his or her output or performance. The most prototypical arc of a star's career begins with increasing performance and a rise to stardom, followed by the prime of one's career, and then followed by a downward trend in productivity at the end of a career. We found this pattern in the performance of NFL star QBs. Researchers focusing on star human capital in the future would be well served to consider where within this career arc the stars are when developing their predictions. This could be an important contingency to the impact of star individuals, as it relates to both peer and organizational performance.

Second, we add to the growing literature on peer effects (Agrawal et al., 2014; Azoulay et al., 2010; Brown, 2011; Guryan, Kroft, & Notowidigdo, 2009; Lacetera et al., 2004; Oettl, 2012). This study examines a specific type of star performer not examined in prior research, a falling star—one who no longer produces at their peak level. These stars are likely to have moved beyond 'lone wolf' or 'all-star' status (Oettl, 2012) and may be moving toward 'helpful' status. Unlike prior studies that have focused only on knowledge sharing mechanisms, this study attempts to tease apart the mechanism by which stars impact their colleagues in a context in which both complementarities and/or knowledge sharing could be present and in job designs that require varying levels of both physical and intellectual abilities. This study found empirical evidence in favor of a complementarity mechanism and found less compelling empirical evidence for a knowledge sharing mechanism. However, qualitative evidence suggests that knowledge sharing is still seen as important in our context and is often referenced as reasoning for adding a falling star. Researchers examining peer effects, even in knowledge-intensive industries, should also examine the role of complementarities, as these could be just as important (or potentially more important) than the simple knowledge-sharing mechanism usually invoked.

This study adds to the frequently cited and growing literature on the mobility of star performers. The literature on star mobility has largely found that adding a star individual leads to lower organizational performance, largely due to having to overcome the adjustment costs of changing organizations (Groysberg et al., 2008; Groysberg & Lee, 2008, 2009). These studies found that 1) the star performs at a lower level than expected in their new organization, and 2) the star appropriates the value they create (Coff, 1999; Groysberg et al., 2008). This study shows that while these prior studies are correct that a star loses some firm-specific human capital, it does not mean that they cannot have a positive impact on their new organization right away. When examining the impact of star performers on organizations, researchers can look beyond the star's individual performance and examine whether the star can

have an impact on the performance of their new colleagues. Essentially, by hiring an 'A' player, the performance of their 'B' players increases (Molloy & Barney, 2015) in a way that the organization can benefit.

This study is not without limitations. First, when studying sports, there are questions surrounding the generalizability of the results. Sports, however, do provide a valuable context for many management studies. It has been argued that "there is sufficient contextual overlap to reasonably expect that knowledge will generalize, regardless of whether the direction is from work (or education, or military, or other organizational context) to sport, or conversely," (Day, Gordon, & Fink, 2012, p. 399). Other researchers also show how sports provide a strong context for organizational researchers (Wolfe et al., 2005) and how professional athletes can be seen as knowledge workers (Chacar & Hesterly, 2008). NFL players, arguably, meet the conditions that Chacar and Coff (2000) use to define knowledge-based employees. It has also been shown that the distribution of wins in sports is closely related to winning in industry competition (Powell, 2003). The NFL context is one in which both physical and mental skills are needed (Gould, 2000), and careers are short. Future research should more closely examine peer effects and potential mechanisms across different organizational contexts. Another potential limitation in studying stars is the small sample size, as star performers are, by definition, rare. Combining this with our context of professional football, where few individuals play in the NFL, we are left with a small sample size. This leads to low power in our statistical tests, especially our robustness checks where the sample is split. This low power might make us miss some important nuances in the way a star

Table 8
Representative quotes and articles referencing falling star quarterback mobility events.

Player	New Team	Quotes	Source
Michael Vick	Eagles	Two days prior to Vick's supposed out pattern, he told reporters, "I'm always going to be a leader on this team, regardless of what my place is."	https://bleacherreport.com/articles/1815872-michael-vick-setting-an-example-for-the-philadelphia-eagles-with-his-leadership
		When training camp opened a month later, he claimed no role was too small. "If I have to watch film with all the quarterbacks, or if I have to coach the quarterbacks one day, then that's what I'll do."	
Kurt Warner	Giants and Cardinals	His most impressive moments were never born from triumph and rarely seen on a football field. When his career bottomed out in New York, Warner remained fully supportive of Tom Coughlin's decision to bench him in favor of Eli Manning. When he was stripped of his starting job in Arizona, he never took his frustrations out on Matt Leinart, continuing to mentor the young quarterback with an open heart.	https://www.azcentral.com/story/sports/nfl/cardinals/2017/08/05/bickley-former-cardinals-quarterback-kurt-warner-wrote-his-own-story/542916001/
Brett Favre	Vikings	I had been around [Steve Hutchinson] and Jared [Allen] and those guys, and I knew how well Brett would add to the locker room. I knew they'd love him. I knew it would be instant. We just kind of watched it on TV. It was kind of the circus had come to town. But more than that, we had an ally in the locker room. It was instant change.	https://www.startribune.com/the-day-brett-favre-became-a-minnesota-viking/523002451/
		We walked up the back steps, and as luck would have it here comes Percy Harvin from the cafeteria, and that was the first person Brett met. Brett said to him, 'Hey, I heard great things about you. Make me look good.' Big smile on Percy's face, of course. He was spoiled playing anywhere in his first year in the NFL with Brett Favre as his quarterback. He ended up being offensive rookie of the year.	
Matt Ryan	Colts	From the moment he came to Indianapolis, teammates and coaches have praised Ryan for his leadership and for keeping everyone in the building accountable. Ryan will routinely stop teammates in the hallway and quiz them about certain plays, where they are supposed to be, and what their assignment is.	https://www.si.com/nfl/colts/news/matt-ryan-leadership-sloppy-colts-practice-stephon-gilmore
Randall Cunningham	Vikings	Brad had better anticipation and less arm strength, but Randall has a bigger arm and could ad lib and do other things that other quarterbacks can't do. We knew that, in having Randall, we had one of the best backups.	https://www.si.com/nfl/2018/06/27/1998-minnesota-vikings-randy-moss-randall-cunningham-cris-carter-dennis-green-gary-anderson
		He comes in, and I can tell he's not really on top of it just yet. Then once he becomes the starter he transforms—his focus and everything. He knew he never had to force the ball. Someone was going to be open. He was so focused on being a pocket quarterback, proving everyone wrong. He wanted to so much to show that he could that he actually wouldn't run the ball. Teams were spying him, waiting for him to take off—and he never did.	
Matt Hasselback	Colts	Working with Luck has been a joy, even if Hasselbeck's thrown only three passes this season and likely won't throw many more. The reward has indeed been giving everything he has to a player who has the job he would love to have again.	https://sports.yahoo.com/news/nfl–matt-hasselbeck-s-mentoring-of-andrew-luck-includes-a-rigged-contest-and-rubbing-people-the-wrong-way-010528025.html
		At 38 he knows a starting job on a good team will not be his. He doesn't want to give up football although he's thinking about the future, even joking that he might want to buy a soccer team with Luck someday. For now he has a role. Maybe the hardest one he's had to play:	

(*continued on next page*)

Table 8 (*continued*)

Player	New Team	Quotes	Source
Mark Brunell	Jets	The invisible supporter of a younger man's rising star. "I'm still a player now," he said. "I'm wearing these shoulder pads. My job out here is that, if and when I'm on the field, whether that's a backup role, wherever it is, my job is to move the team. So that's first and foremost to me.	https://nypost.com/2010/08/03/mentor-job-a-good-fit-for-jets-brunell/
Jeff Garcia	Buccaneers	"At the same time, I think I'm in a position, because of my experience, to help the young guys." "He's unbelievable. I just wish you guys could see the look in his eyes on the field. I know he's taking that out to our huddle. I know he's putting those eyes on those guys in the huddle, and I know it has something to do with the way they're playing," coach Jon Gruden said.	https://www.ocala.com/story/news/2007/09/25/garcias-leadership-boosting-bucs/31220323007/
Drew Bledsoe	Bills	"That's what this franchise needed. We needed a guy that has that kind of grit, that kind of determination. And, he has talent to go with it. You see the great competitor in him. He's never going to quit on a play…. That's one of the reasons that we have a chance." After enduring a bitter quarterback carousel for the past two years – Doug Flutie one week, Rob Johnson the next – free agent signee Drew Bledsoe has finally provided some stability and much-needed leadership to the Buffalo Bills huddle.	https://www.espn.com/nfl/news/2002/0915/1432320.html
Dave Krieg	Chiefs, Lions, Cardinals, Bears and Oilers	"You can tell he's been in this league a long time,'' rookie receiver Josh Reed said. "He was poised the whole time. He was leading us down the field and we just had to follow him.' Krieg may not have been flashy all the time, but he knew how to get the job done, and as many NFL analysts bemoan the mediocre quarterback play that is prevalent through much of the league today, Krieg looks all the better as the years pass on.	https://bleacherreport.com/articles/179614-dave-krieg-a-historical-portrait
Boomer Esiason	Cardinals	Krieg was due to make $1.3 million, and coach Vince Tobin acknowledged the Cardinals would save money while using Esiason as a mentor for Kent Graham and Stoney Case. Last year, the quarterbacks were Krieg, Mike Buck and Case. But Tobin expects Esiason to start and help rebuild a team that collapsed last year under Buddy Ryan. "When you are building an organization, you have to plan into the future, and hopefully Boomer can take us farther into the future than Dave Krieg," Tobin said	https://www.spokesman.com/stories/1996/apr/09/cardinals-sign-esiason-drop-krieg/
Steve McNair	Ravens	"He's the best player I've ever played with," said Ravens cornerback Samari Rolle, who played with McNair for the Titans. "He's a sure-fire Hall of Famer. It hurts because Steve was like a mentor to me." Said linebacker Ray Lewis: "There is no greater warrior or player with a bigger heart than Steve McNair."	https://www.baltimoresun.com/sports/bal-steve-mcnair-0704-story.html
Donovan McNabb	Vikings	The Vikings signed McNabb to a one-year contract, but it is a much bigger investment than one year. He will be the main role model for young Christian Ponder. What Ponder learns under McNabb will follow him for the rest of his career, and that is a great thing. McNabb is model quarterback. He is steady in the pocket, yet he moves when he needs to.	https://bleacherreport.com/articles/795224–5-reasons-why-donovan-mcnabb-was-the-right-choice-for-the-minnesota-vikings
Peyton Manning	Broncos	"He's a guy that raises all boats. He's already made (his teammates) better, and they haven't met him yet just because of the type of person he is, his reputation and what he's done in this league. So, he's just going to have a tremendous effect on the Denver Broncos."	https://www.espn.com/nfl/story/_/id/7713078/2012-nfl-free-agency-peyton-manning-denver-broncos-agree-96 m-deal
Warren Moon	Cheifs	"I think he'll be very helpful to Elvis Grbac in his development," Chiefs head coach Gunther Cunningham said. "I think the players will respond well to him because I think he carries respect throughout the league."	https://www.profootballhistory.com/warren-moon/

interacts with their organization. Future research should examine falling stars in other contexts where there are more overall individuals, such as science or medicine, where a larger sample size is needed to detect smaller effects.

Importantly, the scope of our study suggests that 'falling stars' are particularly prevalent in highly competitive, high pressure, and/or high stress environments (such as professional sports). It is in these contexts that we expect to see that the rapid declining performance of star performers should be particularly salient. This leads to potential future research examining how star performers decline in other settings. For example, in some industries, the declines may be less salient but may still have impacts when these individuals change organizations.

In conclusion, we found that falling stars have a positive impact on the organizations they join. This study found that adding a star has a

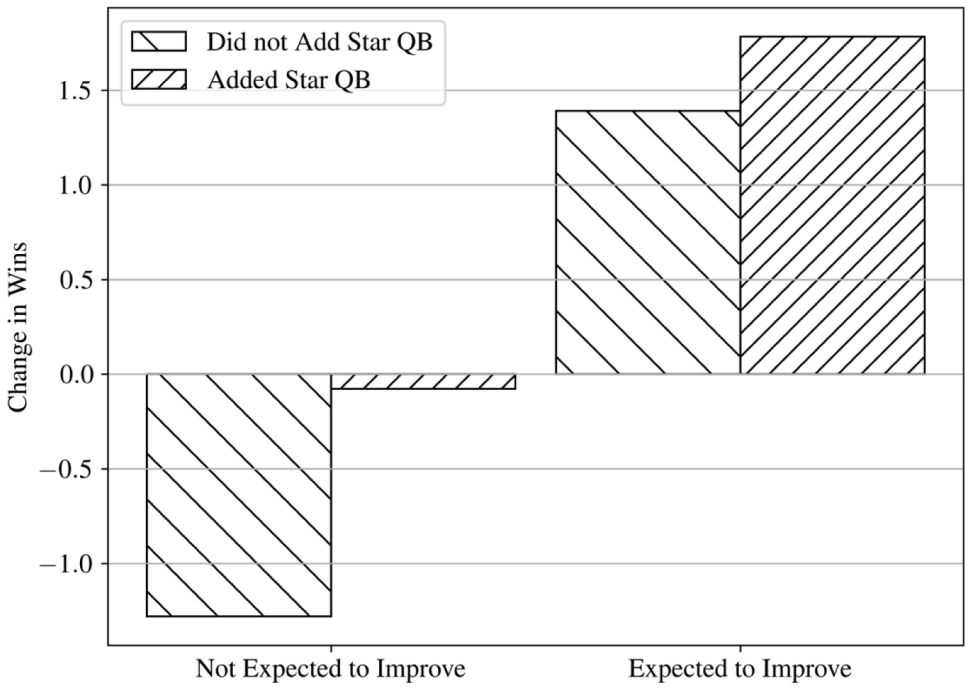

Fig. 2. Change in wins versus expected performance and adding a star QB.

positive impact on the star's new colleagues but that there are several contingencies to this effect. The new colleagues get better only when they have direct interaction with the added star and when they have enough knowledge and skills to benefit. A star does not seem to have any impact on the highest performing or the lowest performing colleagues. Finally, adding a falling star is shown to positively increase the performance of the organization.

CRediT authorship contribution statement

Matthew A. Barlow: Conceptualization, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Methodology, Project administration. **William S. Hesterly:** Conceptualization, Writing – review & editing. **J. Cameron Verhaal:** Conceptualization, Data curation, Writing - review & editing, Investigation.

Declaration of Competing Interest

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References

Abadie, A., & Imbens, G. W. (2011). Bias-corrected matching estimators for average treatment effects. *Journal of Business & Economic Statistics, 29*(1), 1–11. https://doi.org/10.1198/jbes.2009.07333

Agrawal, A. K., McHale, J., & Oettl, A. (2014). Why stars matter. *National Bureau of Economic Research.* http://www.nber.org/papers/w20012.

Aime, F., Johnson, S., Ridge, J. W., & Hill, A. D. (2010). The routine may be stable but the advantage is not: Competitive implications of key employee mobility. *Strategic Management Journal, 31*(1), 75–87. https://doi.org/10.1002/smj.809

Azoulay, P., Graff Zivin, J. S., & Wang, J. (2010). Superstar extinction. *The Quarterly Journal of Economics, 25*, 549–589.

Becker, G. S. (1962). Investments in human capital: A theoretical analysis. *The Journal of Political Economy, 70*(5), 9–49.

Becker, G. S. (1993). *Human capital: A theoretical and empirical analysis with special reference to education* (3rd ed.). The University of Chicago Press.

Berman, S. L., Down, J., & Hill, C. W. L. (2002). Tacit knowledge as a source of competitive advantage in the National Basketball Association. *Academy of Management Journal, 45*(1), 13–31.

Brown, J. (2011). Quitters never win: The (adverse) incentive effects of competing with superstars. *Journal of Political Economy, 119*(5), 982–1013. https://doi.org/10.1086/663306

Call, M. L., Nyberg, A. J., & Thatcher, S. M. B. (2015). Stargazing: An integrative conceptual review, theoretical reconciliation, and extension for star employee research. *Journal of Applied Psychology, 100*(3), 623–640. https://doi.org/10.1037/a0039100

Chacar, A. S., & Coff, R. W. (2000). Deconstructing a knowledge-based advantage: Rent generation, rent appropriation and performance in investment banking. In R. Bresser, D. Heuskel, & R. Nixon (Eds.), *Winning Strategies in a Deconstructing World* (pp. 245–266). Wiley.

Chacar, A. S., & Hesterly, W. (2008). Institutional settings and rent appropriation by knowledge-based employees: The case of Major League Baseball. *Managerial and Decision Economics, 29*(2–3), 117–136. https://doi.org/10.1002/mde.1395

Chambers, E. G., Foulon, M., Hanfield-Jones, H., Hankin, S. M., & Michaels, E. G., III (1998). The war for talent. *The McKinsey Quarterly*, 44–57.

Chan, T. Y., Li, J., & Pierce, L. (2014a). Learning from peers: Knowledge transfer and sales force productivity growth. *Marketing Science, 33*(4), 463–484. https://doi.org/10.1287/mksc.2013.0831

Chan, T. Y., Li, J., & Pierce, L. (2014b). Compensation and peer effects in competing sales teams. *Management Science, 60*(8), 1965–1984. https://doi.org/10.1287/mnsc.2013.1840

Coff, R., & Kryscynski, D. (2011). Invited editorial: Drilling for micro-foundations of human capital-based competitive advantages. *Journal of Management, 37*(5), 1429–1443. https://doi.org/10.1177/0149206310397772

Coff, R. W. (1999). When competitive advantage doesn't lead to performance: The resource-based view and stakeholder bargaining power. *Organization Science, 10*(2), 119–133.

Crook, T. R., Todd, S. Y., Combs, J. G., Woehr, D. J., & Ketchen, D. J., Jr. (2011). Does human capital matter? A meta-analysis of the relationship between human capital and firm performance. *Journal of Applied Psychology, 96*(3), 443–456.

Day, D. V., Gordon, S., & Fink, C. (2012). The sporting life: Exploring organizations through the lens of sport. *The Academy of Management Annals, 6*(1), 397–433.
Ernst, H., Leptien, C., & Vitt, J. (2000). Inventors are not alike: The distribution of patenting output among industrial R&D personnel. *IEEE Transactions on Engineering Management, 47*(2), 184–199.
Ethiraj, S. K., & Garg, P. (2012). The division of gains from complementarities in human-capital-intensive activity. *Organization Science, 23*(3), 725–742.
Felin, T., & Foss, N. J. (2005). Strategic organization: A field in search of micro-foundations. *Strategic Organization, 3*(4), 441–455.
Felin, T., Foss, N. J., Heimeriks, K. H., & Madsen, T. L. (2012). Microfoundations of routines and capabilities: Individuals, processes, and structure: Microfoundations of routines and capabilities. *Journal of Management Studies, 49*(8), 1351–1374. https://doi.org/10.1111/j.1467-6486.2012.01052.x
Felin, T., & Hesterly, W. S. (2007). The knowledge-based view, nested heterogeneity, and new value creation: Philosophical considerations on the locus of knowledge. *Academy of Management Review, 32*(1), 195–218.
Galenson, D. W. (2009). Old masters and young geniuses: The two life cycles of human creativity. *Journal of Applied Economics, 12*(1), 1–9.
Galenson, D. W. (2011). *Old masters and young geniuses: The two life cycles of artistic creativity*. Princeton University Press.
Gould, S. J. (2000). *The Brain of Brawn*. New York Times.
Groysberg, B., & Lee, L.-E. (2008). The effect of colleague quality on top performance: The case of security analysts. *Journal of Organizational Behavior, 29*, 1123–1144.
Groysberg, B., & Lee, L.-E. (2009). Hiring stars and their colleagues: Exploration and exploitation in professional service firms. *Organization Science, 20*(4), 740–758.
Groysberg, B., Lee, L.-E., & Nanda, A. (2008). Can they take it with them? The portability of star knowledge workers' performance. *Management Science, 54*(7), 1213–1230.
Guryan, J., Kroft, K., & Notowidigdo, M. J. (2009). Peer effects in the workplace: Evidence from Random groupings in professional golf tournaments. *American Economic Journal: Applied Economics, 1*(4), 34–68. https://doi.org/10.1257/app.1.4.34
Hasan, S., & Bagde, S. (2013). The mechanics of social capital and academic performance in an Indian college. *American Sociological Review*, 000312413505198.
Hasan, S., & Bagde, S. (2015). Peers and network growth: Evidence from a natural experiment. *Management Science, 61*(10), 2536–2547. https://doi.org/10.1287/mnsc.2014.2109
Hatch, N. W., & Dyer, J. H. (2004). Human capital and learning as a source of sustainable competitive advantage. *Strategic Management Journal, 25*(12), 1155–1178. https://doi.org/10.1002/smj.421
Hitt, M. A., Bierman, L., Shimizu, K., & Kochhar, R. (2001). Direct and moderating effects of human capital on strategy and performance in professional service firms: A resource-based perspective. *Academy of Management Journal, 44*(1), 13–28.
Holcomb, T. R., Holmes, R. M., Jr., & Connelly, B. L. (2009). Making the most of what you have: Managerial ability as a source of resource value creation. *Strategic Management Journal, 30*, 457–485.
Huckman, R. S., & Pisano, G. P. (2006). The firm specificity of individual performance: Evidence from cardiac surgery. *Management Science, 52*(4), 473–488.
Imbens, G. W. (2004). Nonparametric estimation of average treatment effects under exogeneity: A review. *The Review of Economics and Statistics, 86*(1), 4–29.
James, B. (1985). *The Bil James historical abstract*. Villard Books.
Lacetera, N., Cockburn, I. M., & Henderson, R. (2004). Do firms change capabilities by hiring new people? A study of the adoption of science-based drug discovery. In *Advances in Strategic Management* (Vol. 21, pp. 133–159). Elsevier Ltd.
Levin, S. G., & Stephan, P. E. (1991). Research productivity over the life cycle: Evidence for academic scientists. *American Economic Review, 81*(1), 114–132.
Lotka, A. J. (1926). The frequency distribution of scientific productivity. *Journal of Washington Academy of Sciences, 16*(12), 317–323.
Mas, A., & Moretti, E. (2009). Peers at Work. *American Economic Review, 99*(1), 112–145. https://doi.org/10.1257/aer.99.1.112
Moir, A. (2009). When does a football player get old? *The Sports Digest, 17*(4), 1–4.
Molloy, J. C., & Barney, J. B. (2015). Who captures the value created with human capital? A market-based view. *Academy of Management Perspectives, 29*(3), 309–325. https://doi.org/10.5465/amp.2014.0152
Narin, F., & Breitzman, A. (1995). Inventive productivity. *Research Policy, 24*, 507–519.
Oettl, A. (2012). Reconceptualizing stars: Scientist helpfulness and peer performance. *Management Science, 58*(6), 1122–1140.
Powell, T. C. (2003). Varieties of competitive parity. *Strategic Management Journal, 24*(1), 61–86. https://doi.org/10.1002/smj.283
Rider, C.I., Wade, J., Swaminathan, A., & Schwab, A. (2016). Racial disparity in leadership: Performance-reward bias in promotions of national football league coaches. *Available at SSRN 2710398*. http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2710398.
Rosen, S. (1981). The economics of superstars. *The American Economic Review, 71*(5), 845–858.
Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandom studies. *Journal of Educational Psychology, 66*, 688–701.
Rubin, D. B., & Thomas, N. (1996). Matching using estimated propensity scores: Relating theory to practice. *Biometrics, 52*(1), 249. https://doi.org/10.2307/2533160
Simontin, D. K. (1997). Creative productivity: A predictive and explanatory model od career trajectories and landmarks. *Psychological Review, 104*(1), 66–89.
Simontin, D. K. (1999). Talent and its development: An emergenic and epigenetic model. *Psychological Review, 106*(3), 435–457.
Stuart, E. A. (2010). Matching methods for causal inference: A review and a look forward. *Statistical Science, 25*(1), 1–21. https://doi.org/10.1214/09-STS313
Subramony, M., & Chadwick, C. (2021). Too much tenure? Nonlinear effects and moderated influences of unit-level tenure and labor productivity. *Human Resource Management, 60*(3), 363–375. https://doi.org/10.1002/hrm.22023
Way, S. F., Morgan, A. C., Clauset, A., & Larremore, D. B. (2017). The misleading narrative of the canonical faculty productivity trajectory. *Proceedings of the National Academy of Sciences, 114*(44). https://doi.org/10.1073/pnas.1702121114
Wolfe, R. A., Weick, K. E., Usher, J. M., Terborg, J. R., Poppo, L., Murrell, A. J., … Jourdan, J. S. (2005). Sport and organizational studies: Exploring synergy. *Journal of Management Inquiry, 14*, 182–210.
Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). The MIT Press.
Zuckerman, H. (1967). Nobel laureates in science: Patterns of productivity, collaboration, and authorship. *American Sociological Review, 32*(3), 391–403.

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